



Institute of Infrastructure Technology Research and Management

END Semester Evaluation
(Start time: 10.00 a.m. and Submission time: 01.00 p.m.)
B. Tech. (All Branches)
Semester - VI

Date: 25/04/2022

Course Name: IPM
Course Code : HS 225002

Max. Marks: 100

Instructions:

1. Course instructor may penalize the student by deducting the marks, if answer is adjudged to be copied during assessment. This act shall be recorded as UFM (unfair means).
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Section A

Attempt all the questions. Each question carriers 10 marks (10X4 = 40 Marks)

Q.1. Discuss the relevance of Time value of money concept in evaluating feasibility of different infrastructure projects.

Q.2 Deliberate the concept of the Infrastructure Asset Management. What is the relevance of Asset Management in infrastructure management?

Q.2. What are the different risk associated with Infrastructure Projects? How we can mitigate the different risks related to infrastructure project?

Q.5 What kind of legal issues can impact on Infrastructure project? Discuss the legal issues with special reference to Lavasa Project.

Section B

Attempt all the questions. Each question carriers 10 marks (10X3= 30 Marks)

Q.1. Calculate the different cash flow of the project for the capital budgeting based on the following data and calculate the feasibility of the project based on Net Present Value (NPV).

Year	1	2	3	4	5
Unit Sales	50000	100000	100000	70000	50000
Price Per Unit	150	150	150	150	130

- Cost of new plant, equipment and installation – 10 million
- Variable Cost – 80/Unit
- Fixed Costs – 5,00,000 per year
- Working Capital – 100000 to get production started. For each year investment will be 10% of sales in the year
- Straight line depreciation. with no salvage value
- Tax Rate – 34%
- Cost of Capital = 15%



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Q.2 Adani Infrastructure Company is planning to undertake an infrastructure project with initial investment of ₹2, 40,000. Consider the following cash flow from the project and suggest whether Adani Infrastructure should invest at 10%, 15% and 20% discount rate. Why and Why not?

Year	2019	2020	2021	2022
Cash flow	₹ 25,000	₹ 75,000	₹1,50,000	₹ 1,50,000

Q.3 Suppose you are building a toll road with the initial investment of ₹1,00,0000. H (high), M (medium), L (low) correspond to three different traffic volumes based on the state of the economy. Based on the following information decided whether you should invest in this project or not?

State of the economy	Probability	Traffic Volume	Revenue
Good	0.25	H (High)	₹ 1400000
Ok	0.45	M (Medium)	₹ 1200000
Bad	0.3	L (Low)	₹ 500000

	Good	Ok	Bad
H (High)	0.6	0.4	0.1
M (Medium)	0.3	0.45	0.3
L (Low)	0.1	0.15	0.6

Section C

Attempt any three questions. Each question carries 10 marks (10X3= 30 Marks)

Q.1 Discuss the importance and role of Public-Private partnership in the development of Airports in India. Critically evaluate the different issues in bidding process of Mumbai and Delhi airports.

Q.2 Discuss the impact of New Telecom policy on Telecom sector. What is the impact of predatory pricing strategy/behaviour of new entrant Jio and its impact on customers and Airtel? Deliberate the pricing strategy for Airtel to counter Jio and retain its market share and revenue.

Q.3) Discuss the relevance of Hyderabad metro project. What are the different planning and management lessons from Hyderabad Metro Project?

Q.4) Mention the key features and relevance of the Bharatmala Pariyojana project. Discuss the various problems in the Bhartmala project and suggest various strategies to overcome these problems.